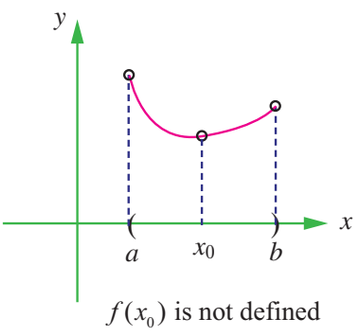
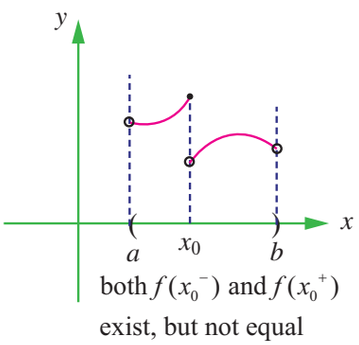
discontinuity. On the other hand, (6) – (9) are in fact discontinuous and really show jumps at certain points.

Students are advised to relate the mathematical definition of continuity corresponds closely with the meaning of the word continuity in everyday language. A continuous process is one that takes place gradually, without interruption or abrupt change. That is, there are no holes, jumps or gaps. Following figure identifies three values of x at which the graph of a function f is not continuous. At all other points in the interval (a, b), the graph of f is uninterrupted and continuous.



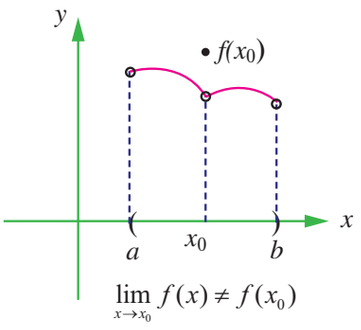
$f(x_0)$ is not defined

Fig. 9.33



both $f(x_0^-)$ and $f(x_0^+)$ exist, but not equal

Fig. 9.34



$\lim_{x \to x_0} f(x) \neq f(x_0)$

Fig. 9.35

Looking at the above graphs (9.33 to 9.35), three conditions exist for which the graph of f is not continuous at $x = x_0$.

It appears that continuity at $x = x_0$ can be destroyed by any one of the following three conditions:

(1) The function is not defined at $x = x_0$.

(2) The limit of $f(x)$ does not exist at $x = x_0$.

(3) The limit of $f(x)$ exists at $x = x_0$, but, it is not equal to $f(x_0)$.

Now let us look at the illustrative examples

Illustration 9.6

(i) $f(x) = x^2+3$ (ii) $f(x) = \dfrac{16-x^2}{4+x}$

(i) As $x \to 2$, the one sided limits are

$$\lim_{x \to 2^-} f(x) = 7$$

$$\lim_{x \to 2^+} f(x) = 7$$

and hence $\lim_{x \to 2} f(x) = 7$ and moreover $f(2)$ is defined and $f(2) = 7 = \lim_{x \to 2} f(x)$. In this case $f(x)$ is continuous at $x = 2$.

(ii) The one sided limits are : $\lim_{x \to -4^-} f(x) = 8$

$$\lim_{x \to -4^+} f(x) = 8$$

and therefore $\lim_{x \to -4} f(x) = 8$ but $f(-4)$ does not exist.

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